MEMORANDUM | KESTREL HOME APPLIANCES

TO: Ritu Deshpande, Head of D2C Operations
FROM: Predictive Analytics & Operations Lead
DATE: October 1, 2026
SUBJECT: Returns Risk Model: Executive Recommendations & Operational Decision Framework

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Executive Summary: Returns cost Kestrel **Rs. 11.04 Lakhs annually** (1,200 returns across 10,504 orders at Rs. 1,150 all-in cost per return). Holding orders with >24h dispatch delay causes a **12% customer cancellation rate** on legitimate orders and directly alienates top-tier **Kestrel Shield VIPs** (who return 2x more often under the free 30-day policy). A blanket hold policy increases net loss by Rs. 41k/year. Instead, our **Smart Pre-Dispatch Confirmation System** recovers **+Rs. 1.41 Lakhs/year in net profit (120.5% ROI)** by preventing 225 returns annually with zero order cancellations.

1. The Decision: Confirmation Calls vs. Blanket Holds

Strategy	Operational Action	Customer Impact	Annual P&L; Impact
Status Quo	Ship all orders blindly	11.4% return rate absorbed as loss	-Rs. 11,04,000
Naive Hold	Hold dispatch on flagged orders (>24h)	12% cancellation of good sales; Shield friction	-Rs. 11,45,160 (Loss!)
Smart Call (Ours)	Auto-dispatch low risk; Rs. 45 call for score ≥ 11.18%	Zero cancellations; proactive white-glove setup	**+Rs. 1,41,304 (Net Gain)**

2. The Numbers: Risk Drivers & Validation

Audited on forward holdout history (April–June 2026), our model achieved **ROC-AUC 0.7869**, **PR-AUC 0.4150** (3.6x over baseline), and **89.6% accuracy** with zero target leakage:

- **Cash on Delivery (COD):** 1.8x higher return likelihood than Prepaid UPI/Cards (18.8% vs. 7.6%).
- **Customer Return History:** Customers with ≥ 2 prior returns have a **40.1% to 100% return rate**.
- **Promised Delivery SLA:** Orders promised at 7–11 days suffer a **15.4%–31.3% return rate** due to buyer remorse.
- **Product Complexity:** Robot Vacuums (19.3%) and Water Purifiers (14.8%) have highest return rates.

3. The Rupees: Unit Economics & Cash ROI

- **Return Cost:** Rs. 1,150 (reverse pickup, inspection, repack, write-down).
- **Confirmation Call Cost:** Rs. 45 per completed outsourced call (prevents 35% of returns on called orders).
- **Break-Even Threshold:** P(return) × (35% × Rs. 1,150) > Rs. 45 $\Rightarrow$ **Score ≥ 11.18%**.
- **Annual Return Recovery:** 2,605 calls placed (Rs. 1.17L cost) → **225 returns prevented (Rs. 2.59L gross savings)** → **Net Bottom-Line Gain: +Rs. 1,41,304/yr (120.5% ROI)**.

4. Action Plan for Next Week

1. **Deploy Smart Dispatch in Warehouse (Monday):** Score < 11.18% → Auto-dispatch immediately. Score ≥ 11.18% → Trigger Rs. 45 confirmation call.
2. **Implement VIP Shield Protocol (Tuesday):** Never hold Shield orders; route flagged VIPs to Meenal's desk for white-glove setup calls.
3. **Shorten Delivery SLAs for High-Risk Pincodes (Wednesday):** Reduce promised delivery SLA from 8 days to ≤ 4 days in top metro clusters.
4. **Board Presentation (Friday):** Present the Rs. 1.41 Lakh net profit recovery backed by our live verified service.